



**Escola Superior
de Tecnologia
e Gestão**

Politécnico de Coimbra

Inteligência Artificial

Nonparametric Tests

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- **SPSS**

Inferential Statistics

- **Types of Tests**

- **Parametric statistical tests**

- makes assumptions regarding the population parameters and distributions
 - main assumption that these tests make is that the values are approximately **normally distributed** in the population

- **Nonparametric statistical tests**

- does **not assume anything about the population parameters.**
 - often called “distribution-free tests” since they do not assume an underlying distribution

Inferential Statistics

- Types of Tests

	Parametric	Nonparametric
Describe one group	Mean SD; SEM; 95%CI	Median Range
Compare one group to a hypothetical value	1-sample t-test	Wilcoxon test (or Chi-square for two possible outcomes)
Compare two unpaired groups	2-sample unpaired t-test	Mann-Whitney test
Compare two paired groups	2-sample paired t-test	Wilcoxon test
Compare three or more unmatched groups	One-Way ANOVA	Kruskal-Wallis test ((or Chi-square for two possible outcomes)
<i>Predict value from measured values</i>	<i>Linear or nonlinear regression</i>	<i>Nonparametric regression</i>

<https://biocomm.eu/2019/01/10/guide-to-essential-biostatistics-i-the-scientific-method-3-2-2-2-2-2-2-2-2/>

Inferential Statistics

- **Types of Tests**

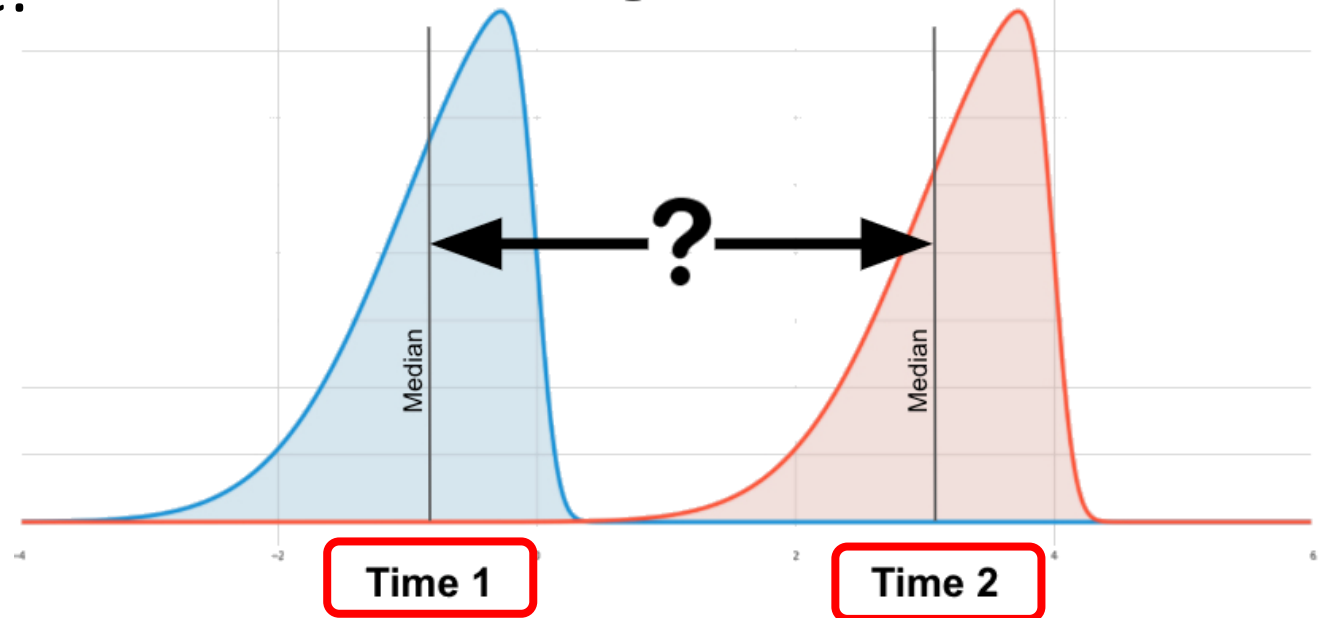
- **Reasons to Use Nonparametric Tests:**

- Nonparametric tests are valid when data does not meet the assumptions associated with the desired parametric test
 - Nonparametric tests are often a good option for small sample sizes where parametric assumptions of normality are worrisome
 - Nonparametric tests can be used when an area of study is best represented by the median (e.g. income)
 - Nonparametric tests can be used to analyse ordinal data, ranked data, or when outliers can't be removed from the data

Inferential Statistics

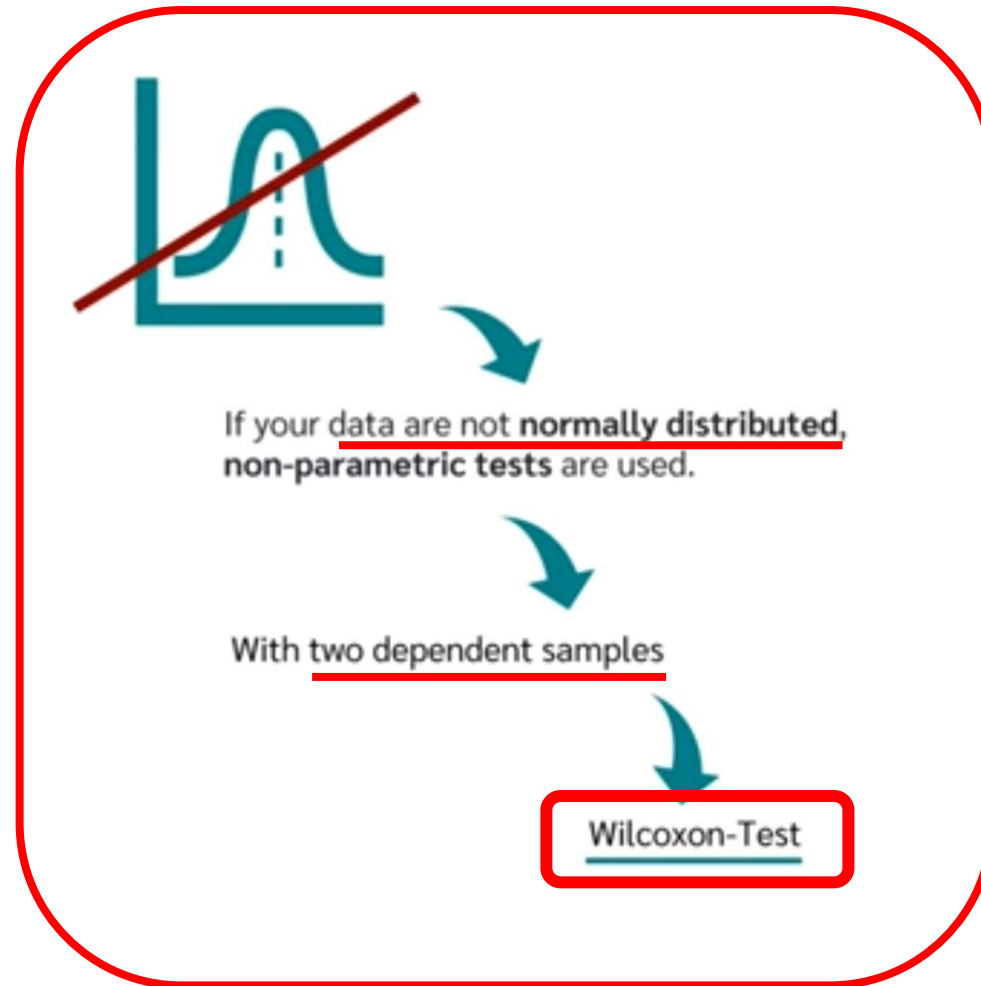
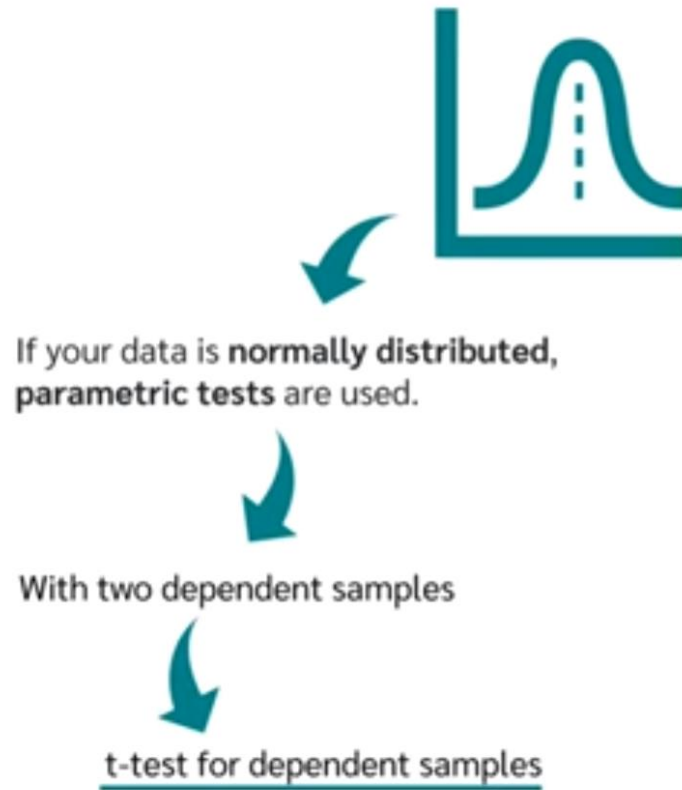
- Nonparametric tests
- Wilcoxon test
- it serves the same purpose as the:
 - **one-sample Student's t-test**
 - **paired Student's t-test**
- unlike the Student's t-test, does not assume that the differences between paired samples are normally distributed.

Wilcoxon Signed Rank Test



Inferential Statistics

- Nonparametric tests
- Wilcoxon test



Inferential Statistics

- Nonparametric tests
- Wilcoxon test

Ranks

		N	Mean Rank	Sum of Ranks
Median - Age (years)	Negative Ranks	33 ^a	37,47	1236,50
	Positive Ranks	40 ^b	36,61	1464,50
	Ties	5 ^c		
	Total	78		

Reaction time morning	Reaction time evening
34	45
36	33
41	35
39	43
44	42
37	42

diff (morning - evening)	Ranking from diff
-11	6 (-)
3	2
6	5
-4	3 (-)
2	1
-5	4 (-)
T ⁺ = 8 & T ⁻ = 13	

<https://datatab.net/tutorial/wilcoxon-test>

In each case the sum of positive and negative rankings

Inferential Statistics

- **Nonparametric tests**
- **Mann-Whitney test**
- it serves the same purpose as the:
 - **unpaired Student's t-test**
- unlike the Student's t-test, does not assume that the differences between unpaired samples are normally distributed

A nominal or ordinal variable with two expressions



Example:

Gender
1 = male
2 = female

Medication
1 = Drug
2 = Placebo

Production facilities
1 = A
2 = B

Independent variable

A metric or ordinal variable



Salary

Wellbeing

Weight

Dependent variable

Inferential Statistics

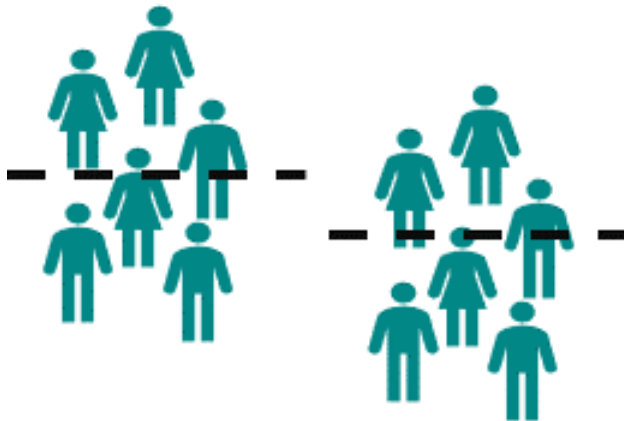
- **Nonparametric tests**
- **Mann-Whitney test**

Hypotheses Mann-Whitney U Tests

For the Mann-Whitney U test, the null hypothesis is that there is no difference in the sum of the rankings in the two groups in the population. The alternative hypothesis is that there is a difference in the sum of the ranks.

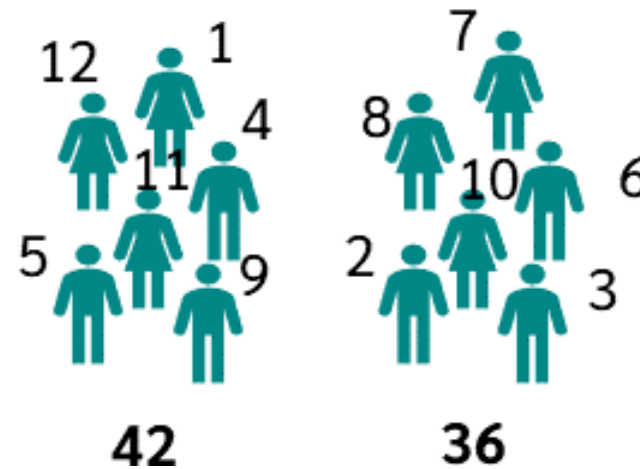
t-Test

Is there a difference in mean?



Mann-Whitney U Test

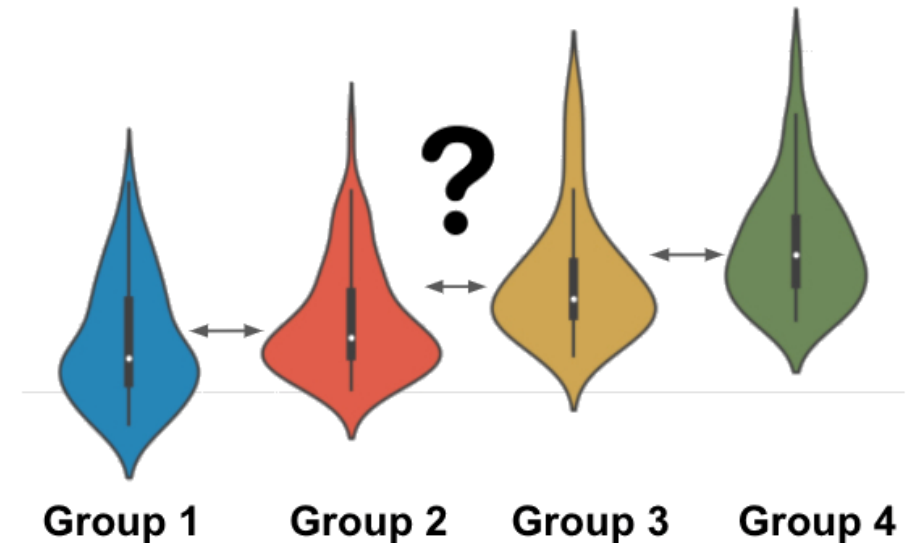
Is there a difference in the rank sum?



Inferential Statistics

- Nonparametric tests
- Kruskal-Wallis test (aka one-way ANOVA on ranks, H test)
- used for comparing two or more independent samples of equal or different sample sizes:
 - it extends the Mann–Whitney U test, which is used for comparing only two groups
- the parametric equivalent of the Kruskal–Wallis test is the one-way analysis of variance (ANOVA)

Kruskal-Wallis One-Way ANOVA



Inferential Statistics

- Nonparametric tests
- Kruskal-Wallis test

Hypothesis Test Summary

	Null Hypothesis	Test	Sig.	Decision
1	The distribution of Overall, how satisfied are you with your life nowadays? is the same across categories of Grouped Age.	Independent-Samples Kruskal-Wallis Test	.000	Reject the null hypothesis.

Asymptotic significances are displayed. The significance level is .05.

Total N	2,042
Test Statistic	44.520
Degrees of Freedom	5
Asymptotic Sig. (2-sided test)	.000

Independent-Samples Kruskal-Wallis Test



Kruskal-Wallis test: H_0 = the median ranks of the groups are the same
 $\text{sig} < 0,05 \rightarrow$ rejectar $H_0 \rightarrow$ the median ranks of the groups are different

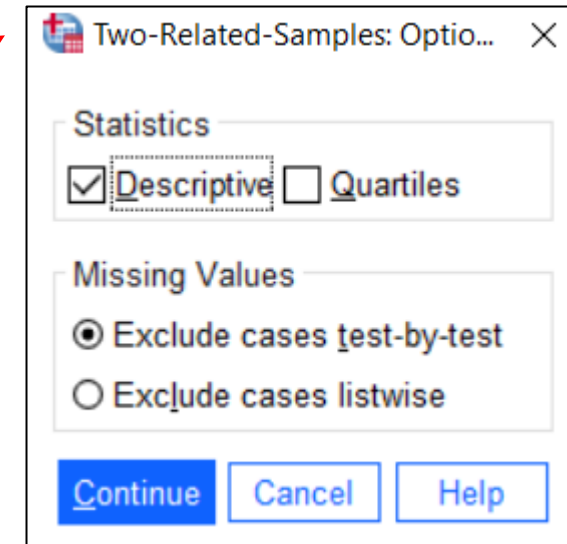
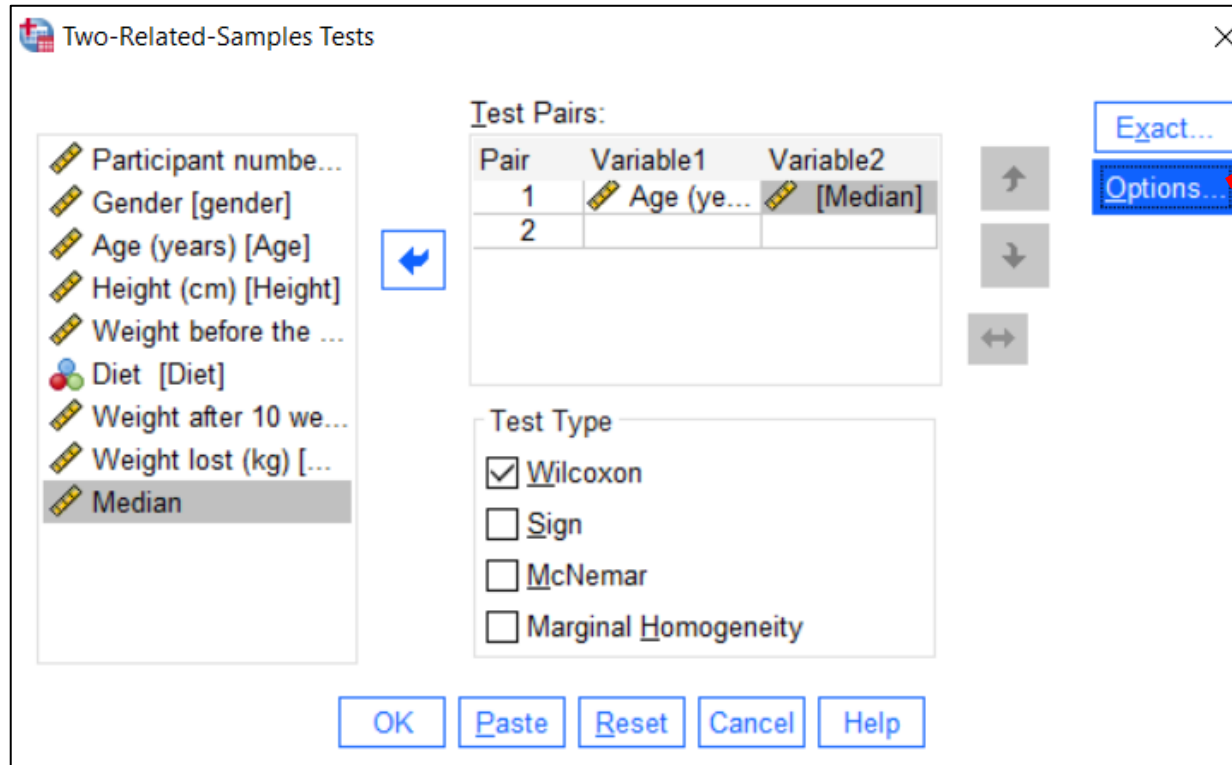
- Transform » Compute Variable

[illegible]

Inferential Statistics

- **SPSS**
- **Wilcoxon test**

Analyse » Nonparametric tests »
Legacy Dialogs » 2 Related Samples



Inferential Statistics

- SPSS
- Wilcoxon test

Analyse » Nonparametric tests »
Legacy Dialogs » 2 Related Samples

Descriptive Statistics					
	N	Mean	Std. Deviation	Minimum	Maximum
Age (years)	78	39,15	9,815	16	60
Median	78	40,0000	,00000	40,00	40,00

Test Statistics^a

Median - Age (years)	
Z	-,627 ^b
Asymp. Sig. (2-tailed)	,531

a. Wilcoxon Signed Ranks Test

b. Based on negative ranks.

Ranks		N	Mean Rank	Sum of Ranks
Median - Age (years)	Negative Ranks	33 ^a	37,47	1236,50
	Positive Ranks	40 ^b	36,61	1464,50
	Ties	5 ^c		
	Total	78		

a. Median < Age (years)

b. Median > Age (years)

c. Median = Age (years)

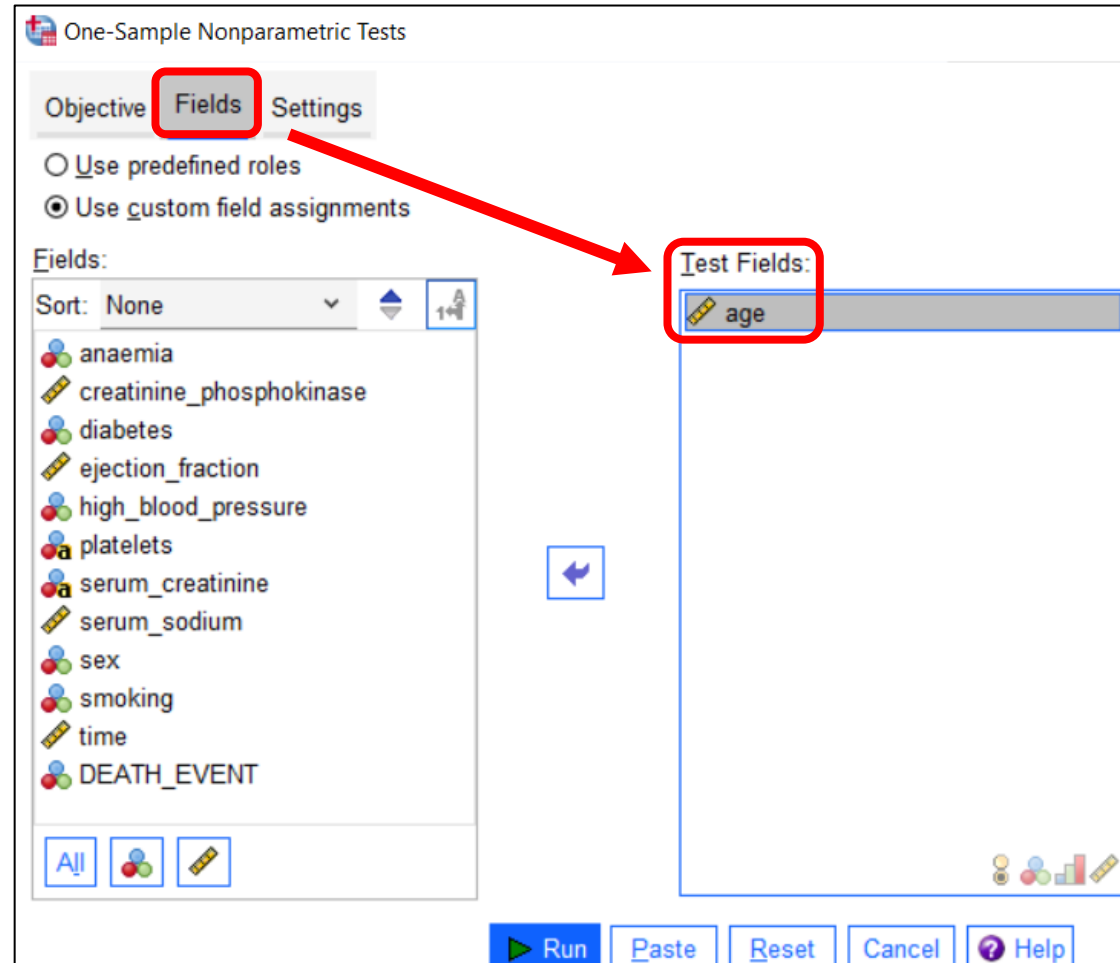
Wilcoxon test: H_0 = grupos possuem medianas iguais
sig > 0,05 -> aceitar H_0 -> grupos possuem medianas iguais, i.e. , a mediana de idades não é significativamente diferente de 40 anos

Inferential Statistics

- **SPSS**

Analyse » Nonparametric tests » One Sample

- **Wilcoxon test (alternative solution)**

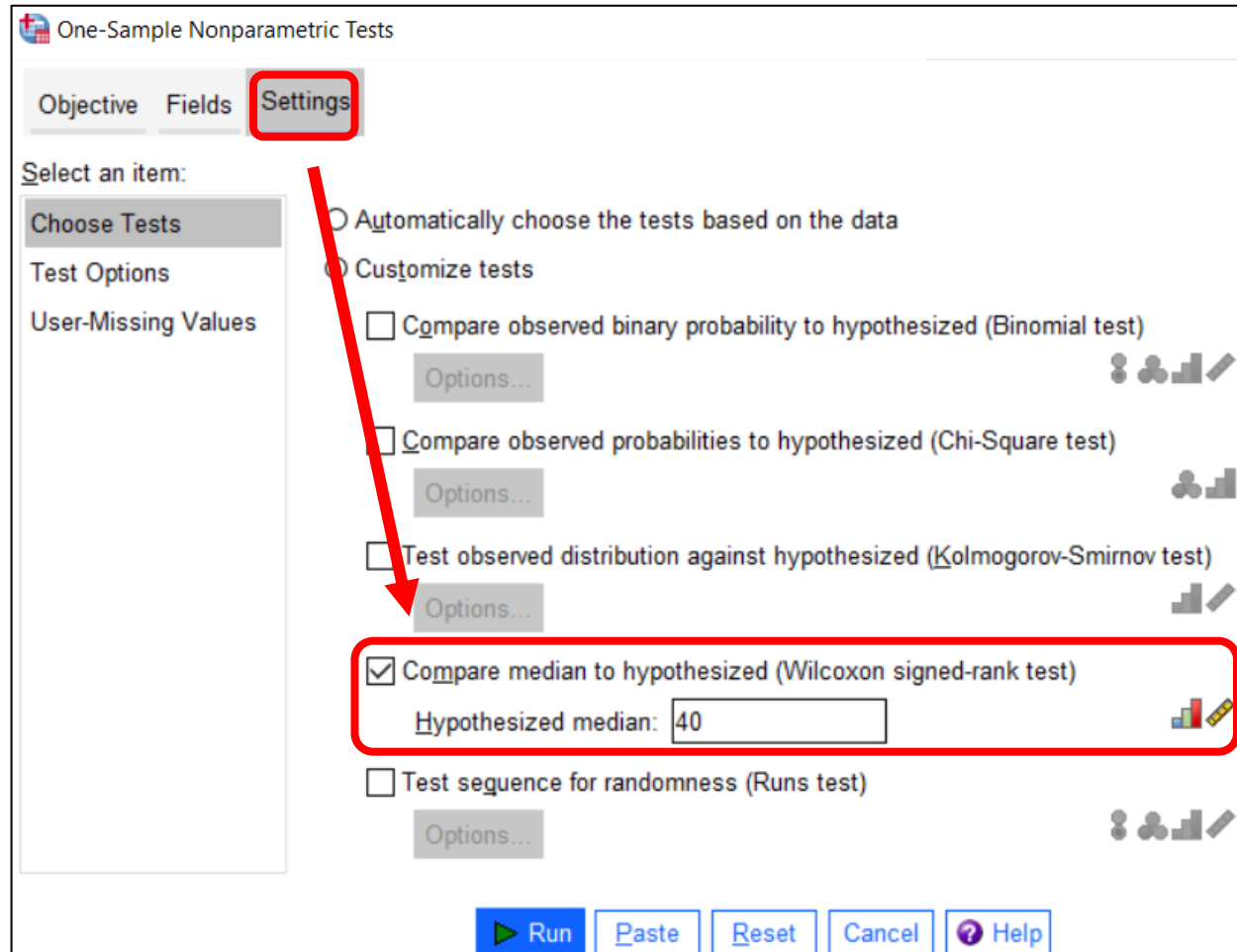


Inferential Statistics

- **SPSS**

Analyse » Nonparametric tests » One Sample

- **Wilcoxon test (alternative solution)**



Inferential Statistics

- **SPSS**

Analyse » Nonparametric tests » One Sample

- **Wilcoxon test (alternative solution)**

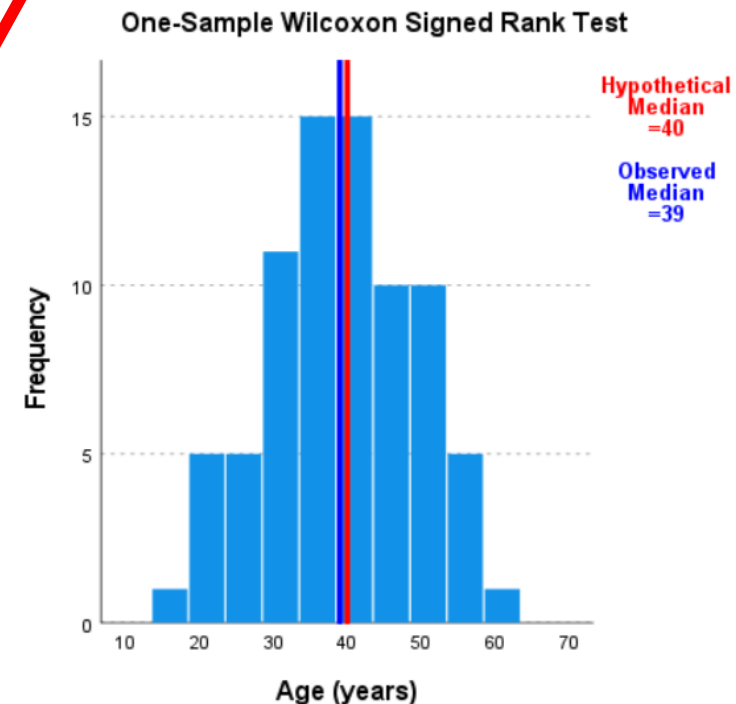
Hypothesis Test Summary			
	Null Hypothesis	Test	Sig. ^{a,b}
1	The median of Age (years) equals 40.	One-Sample Wilcoxon Signed Rank Test	,531
			Decision
			Retain the null hypothesis.

a. The significance level is ,050.

b. Asymptotic significance is displayed.

One-Sample Wilcoxon Signed Rank Test Summary

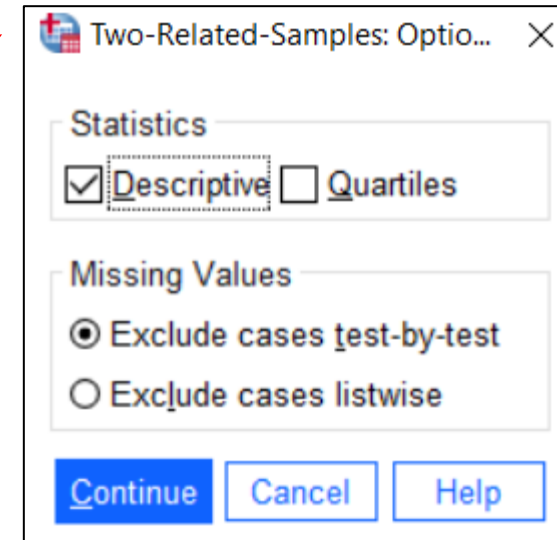
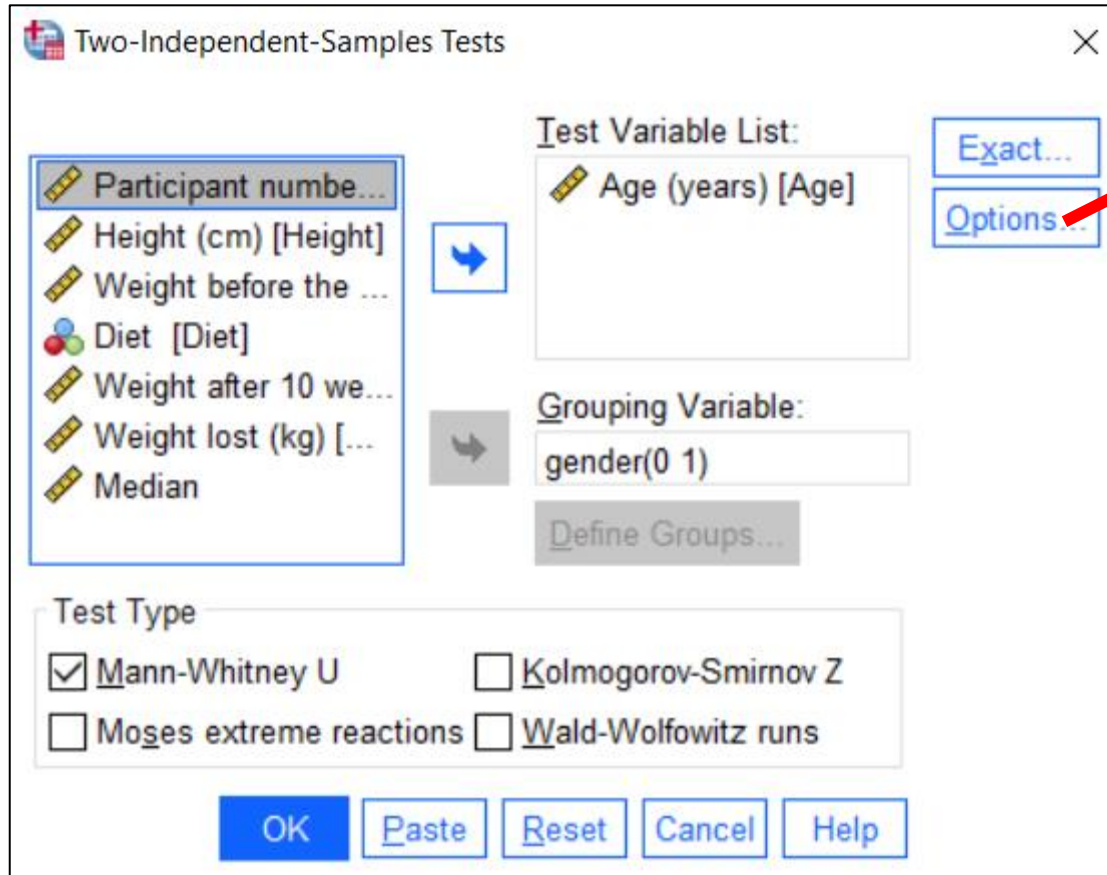
Total N	78
Test Statistic	1236,500
Standard Error	181,756
Standardized Test Statistic	-,627
Asymptotic Sig.(2-sided test)	,531



Inferential Statistics

- **SPSS**
- **Mann-Whitney test**

Analyse » Nonparametric tests » Legacy Dialogs »
2 Independent Samples



Inferential Statistics

- **SPSS**
- **Mann-Whitney test**

Analyse » Nonparametric tests » Legacy Dialogs »
2 Independent Samples

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum
Age (years)	78	39,15	9,815	16	60
Gender	76	,43	,499	0	1

Ranks

	Gender	N	Mean Rank	Sum of Ranks
Age (years)	Female	43	38,41	1651,50
	Male	33	38,62	1274,50
	Total	76		

Test Statistics^a

	Age (years)
Mann-Whitney U	705,500
Wilcoxon W	1651,500
Z	-,042
Asymp. Sig. (2-tailed)	,967

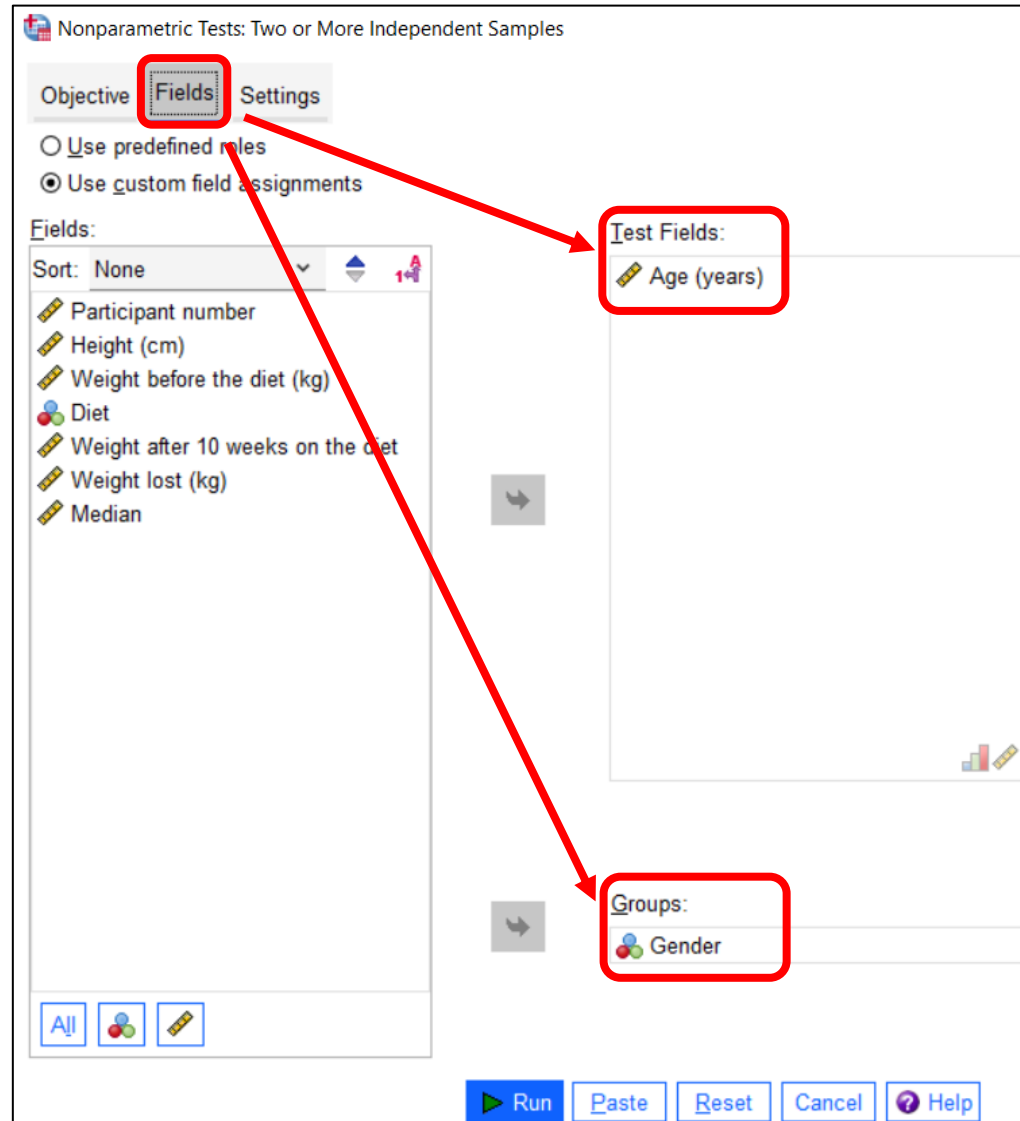
a. Grouping Variable: Gender

Mann-Whitney test: H_0 = grupos possuem medianas iguais
sig > 0,05 -> aceitar H_0 -> grupos possuem medianas iguais, i.e. , a mediana de idades não é significativamente diferente entre os dois grupos (masculino e feminino)

Inferential Statistics

- **SPSS**
- **Mann-Whitney test
(alternative solution)**

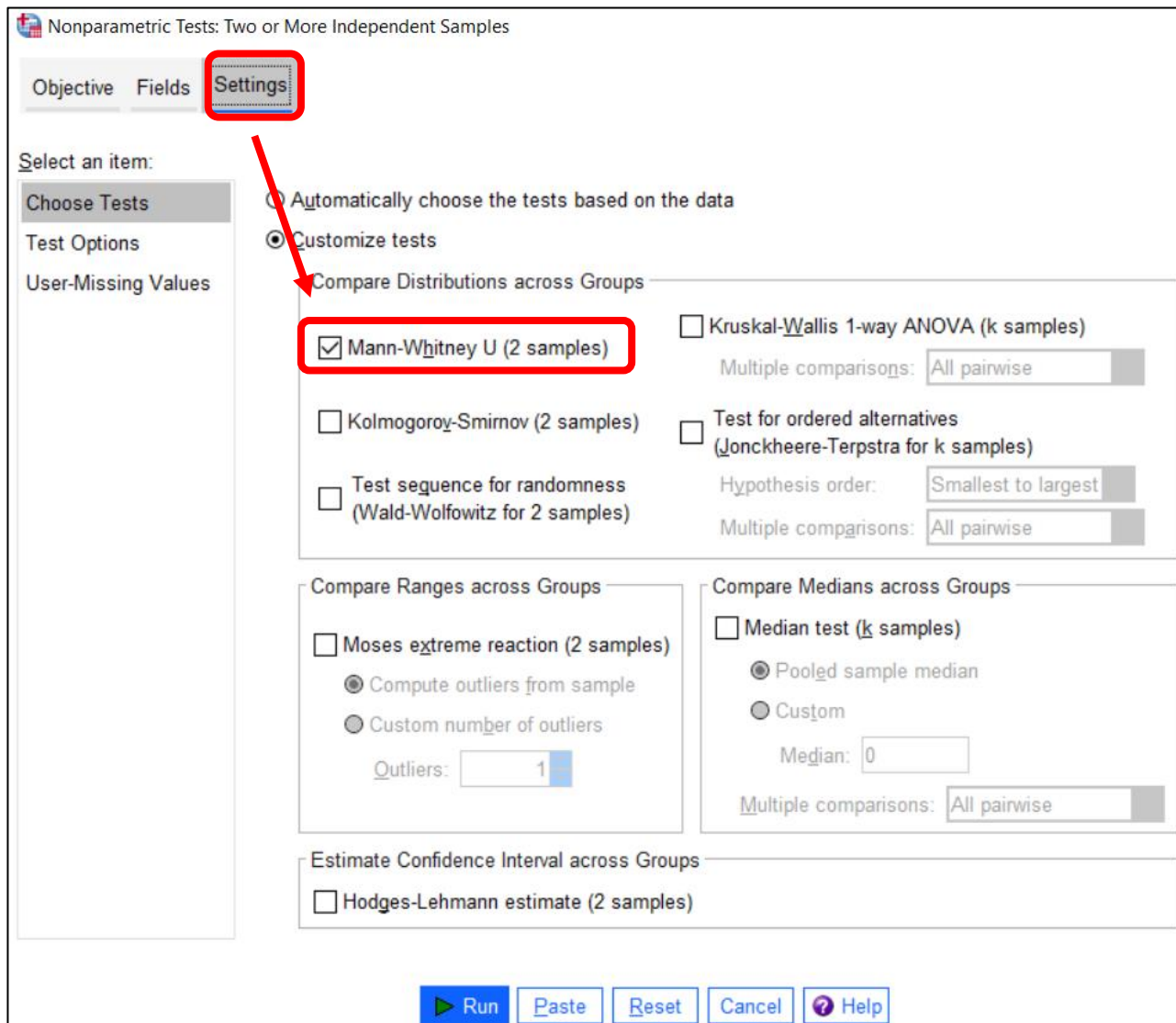
Analyse » Nonparametric tests » Independent Samples



Inferential Statistics

- SPSS
- Mann-Whitney test
(alternative solution)

Analyse » Nonparametric tests » Independent Samples



The image shows the 'Nonparametric Tests: Two or More Independent Samples' dialog box in SPSS. The 'Settings' tab is selected and highlighted with a red box. A red arrow points from the 'Settings' tab to the 'Mann-Whitney U (2 samples)' checkbox, which is also highlighted with a red box. The 'Choose Tests' section on the left lists 'Choose Tests', 'Test Options', and 'User-Missing Values'. The 'Test Options' section is expanded, showing the following options:

- ☒ Automatically choose the tests based on the data
- ☒ Customize tests

Under 'Compare Distributions across Groups':

- ☒ Mann-Whitney U (2 samples)
- ☐ Kruskal-Wallis 1-way ANOVA (k samples)
- ☐ Kolmogorov-Smirnov (2 samples)
- ☐ Test for ordered alternatives (Jonckheere-Terpstra for k samples)
- ☐ Test sequence for randomness (Wald-Wolfowitz for 2 samples)

Under 'Compare Ranges across Groups':

- ☐ Moses extreme reaction (2 samples)
- ☒ Compute outliers from sample
- ☐ Custom number of outliers
- Outliers:

Under 'Compare Medians across Groups':

- ☐ Median test (k samples)
- ☒ Pooled sample median
- ☐ Custom
- Median:

Under 'Estimate Confidence Interval across Groups':

- ☐ Hodges-Lehmann estimate (2 samples)

At the bottom, there are buttons for 'Run', 'Paste', 'Reset', 'Cancel', and 'Help'.

Inferential Statistics

- SPSS
- Mann-Whitney test
(alternative solution)

Analyse » Nonparametric tests » Independent Samples

Hypothesis Test Summary			
	Null Hypothesis	Test	Sig. ^{a,b}
1	The distribution of Age (years) is the same across categories of Gender.	Independent-Samples Mann-Whitney U Test	,967
Decision			
Retain the null hypothesis.			

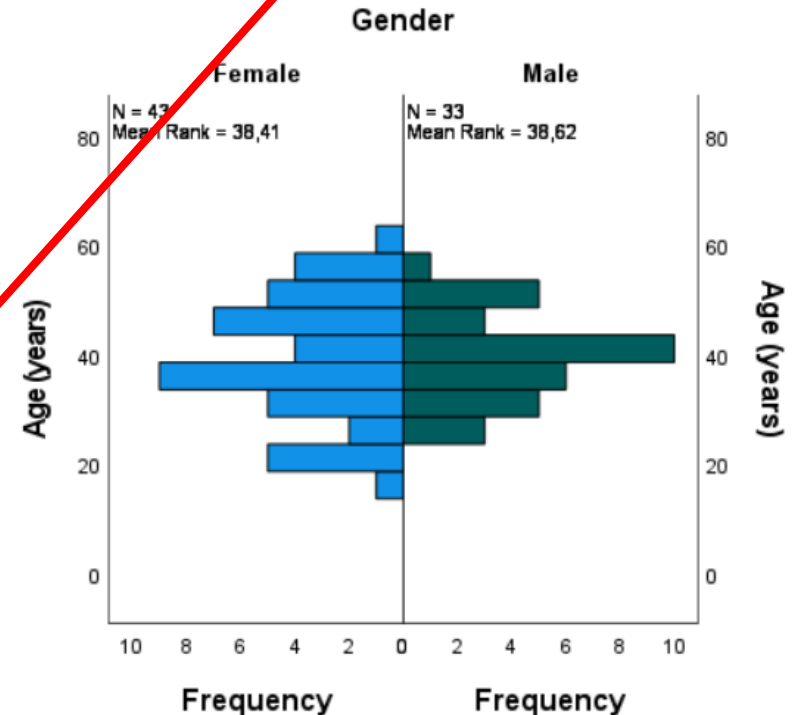
a. The significance level is ,050.

b. Asymptotic significance is displayed.

**Independent-Samples Mann-Whitney U
Test Summary**

Total N	76
Mann-Whitney U	713,500
Wilcoxon W	1274,500
Test Statistic	713,500
Standard Error	95,303
Standardized Test Statistic	,042
Asymptotic Sig.(2-sided test)	,967

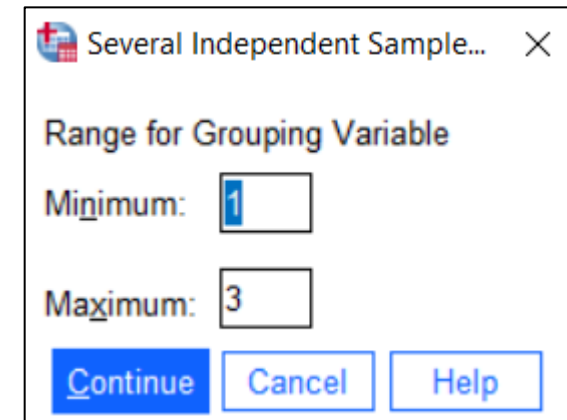
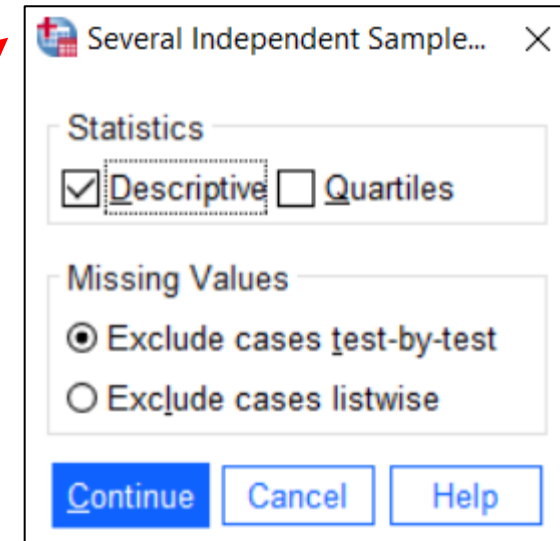
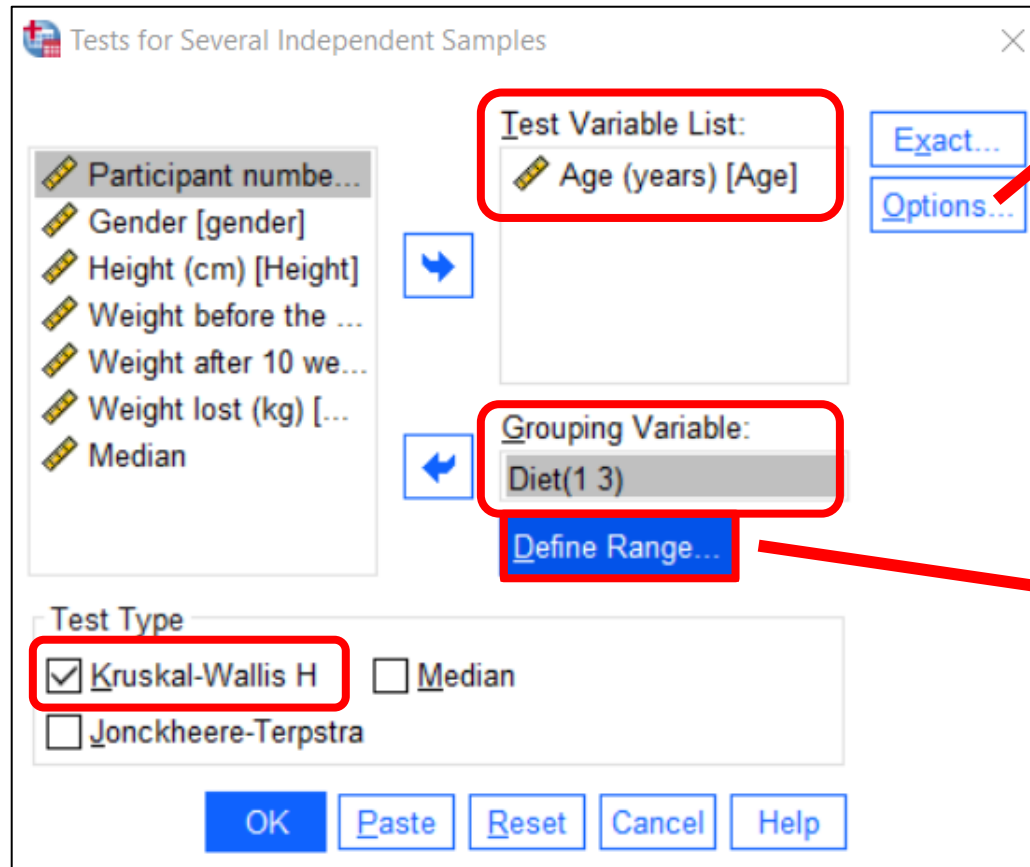
Independent-Samples Mann-Whitney U Test



Inferential Statistics

- SPSS
- Kruskal-Wallis test

Analyse » Nonparametric tests » Legacy Dialogs »
K Independent Samples



Inferential Statistics

- SPSS
- Kruskal-Wallis test

Analyse » Nonparametric tests » Legacy Dialogs »
K Independent Samples

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum
Age (years)	78	39,15	9,815	16	60
Diet	78	2,04	,813	1	3

Ranks

	Diet	N	Mean Rank
Age (years)	1	24	43,56
	2	27	39,87
	3	27	35,52
	Total	78	

Test Statistics^{a,b}

Age (years)	
Kruskal-Wallis H	1,616
df	2
Asymp. Sig.	,446

a. Kruskal Wallis Test

b. Grouping Variable: Diet

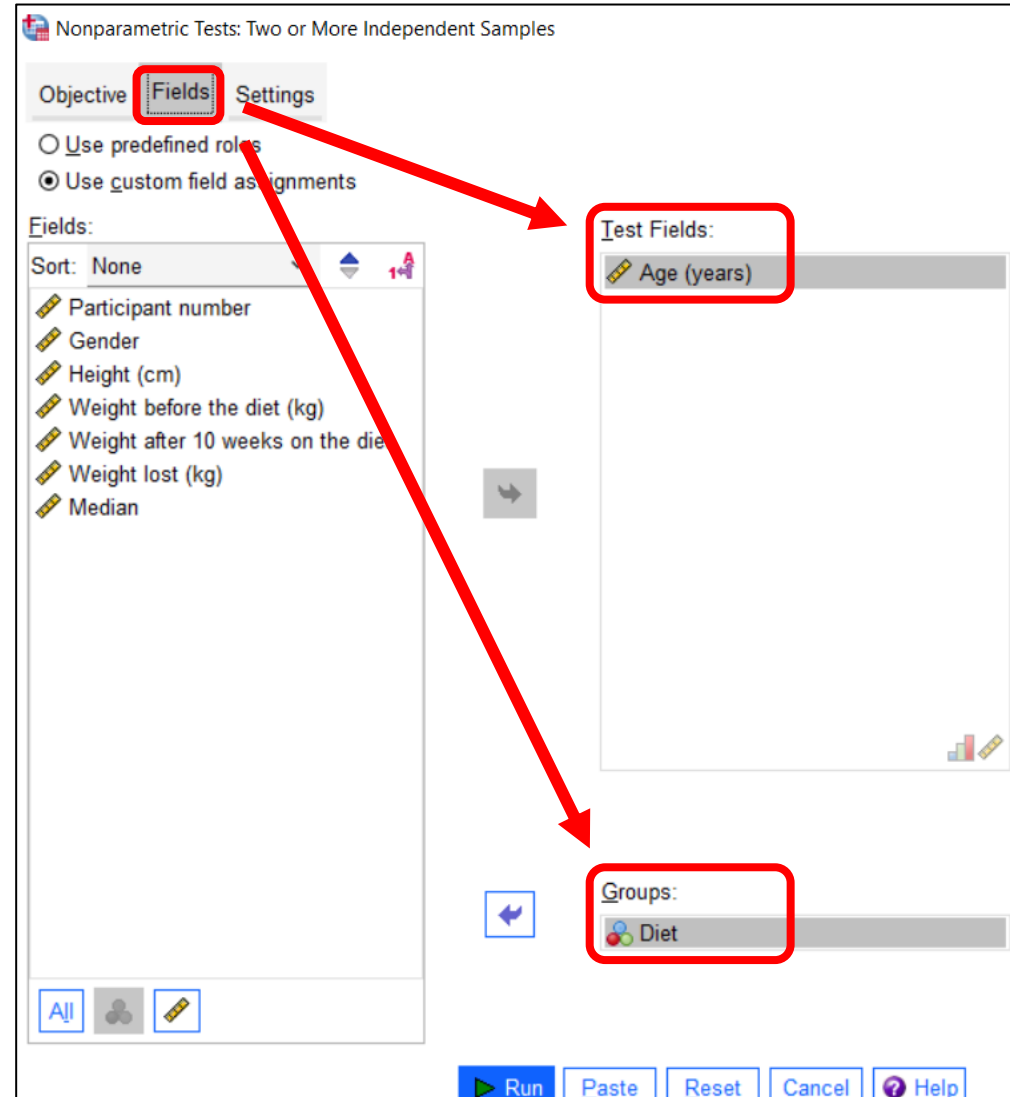
Kruskal-Wallis test: H_0 = the median ranks of the groups are the same
sig > 0,05 -> aceitar H_0 -> the median ranks of the groups are the same

Report: A Kruskal-Wallis H test showed that there was not a statistically significant difference in age between the different diet prescriptions, $H(2) = 1,616$, $p = 0.446$, with a mean rank pain score of 43, 56 for Diet 1 A, 39,87 for Diet 2 and 35,52 for Diet 3.

Inferential Statistics

- **SPSS**
- **Kruskal-Wallis test**
(alternative solution)

Analyse » Nonparametric tests » Independent Samples



Inferential Statistics

- SPSS
- Kruskal-Wallis test
(alternative solution)

Analyse » Nonparametric tests » Independent Samples

Nonparametric Tests: Two or More Independent Samples

Objective Fields **Settings**

Select an item:
Choose Tests
Test Options
User-Missing Values

☐ Automatically choose the tests based on the data
☒ Customize tests

Compare Distributions across Groups

☐ Mann-Whitney U (2 samples)
☐ Kolmogorov-Smirnov (2 samples)
☐ Test sequence for randomness (Wald-Wolfowitz for 2 samples)

☒ **Kruskal-Wallis 1-way ANOVA (k samples)**
Multiple comparisons: All pairwise

☐ Test for ordered alternatives (Jonckheere-Terpstra for k samples)
Hypothesis order: Smallest to largest
Multiple comparisons: All pairwise

Compare Ranges across Groups

☐ Moses extreme reaction (2 samples)
☒ Compute outliers from sample
☐ Custom number of outliers
Outliers: 1

Compare Medians across Groups

☐ Median test (k samples)
☒ Pooled sample median
☐ Custom
Median: 0
Multiple comparisons: All pairwise

Estimate Confidence Interval across Groups

☐ Hodges-Lehmann estimate (2 samples)

Run Paste Reset Cancel Help

Inferential Statistics

- SPSS
- Kruskal-Wallis test
(alternative solution)

Analyse » Nonparametric tests » Independent Samples

Hypothesis Test Summary			
	Null Hypothesis	Test	Sig. ^{a,b}
1	The distribution of Age (years) is the same across categories of Diet.	Independent-Samples Kruskal-Wallis Test	,446
			Decision
			Retain the null hypothesis.

a. The significance level is ,050.

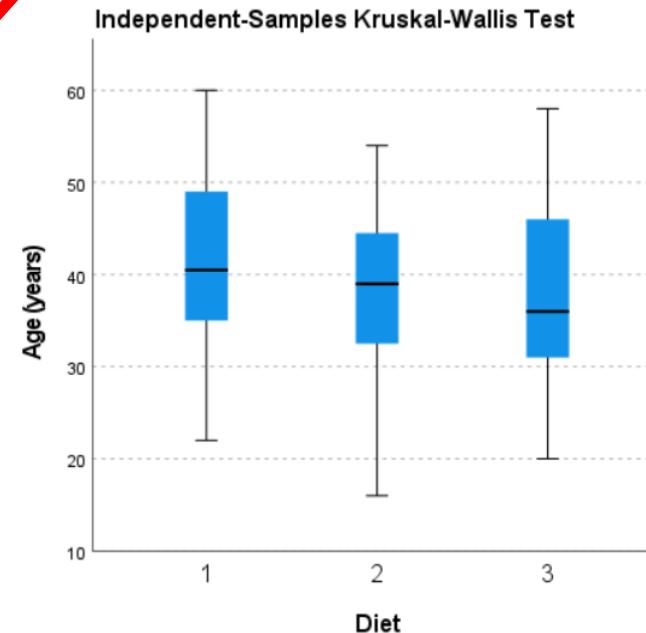
b. Asymptotic significance is displayed.

Independent-Samples Kruskal-Wallis Test Summary

Total N	78
Test Statistic	1,616 ^{a,b}
Degree Of Freedom	2
Asymptotic Sig.(2-sided test)	,446

a. The test statistic is adjusted for ties.

b. Multiple comparisons are not performed because the overall test does not show significant differences across samples.



Bibliography

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- Geron, A. (2019). Hands-on machine learning with scikit-learn, keras, and tensorflow. O'Reilly.
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